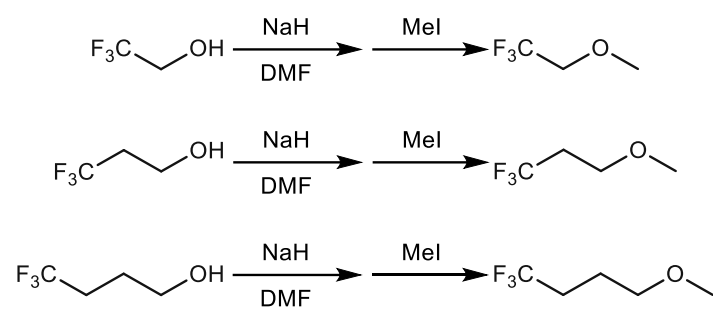
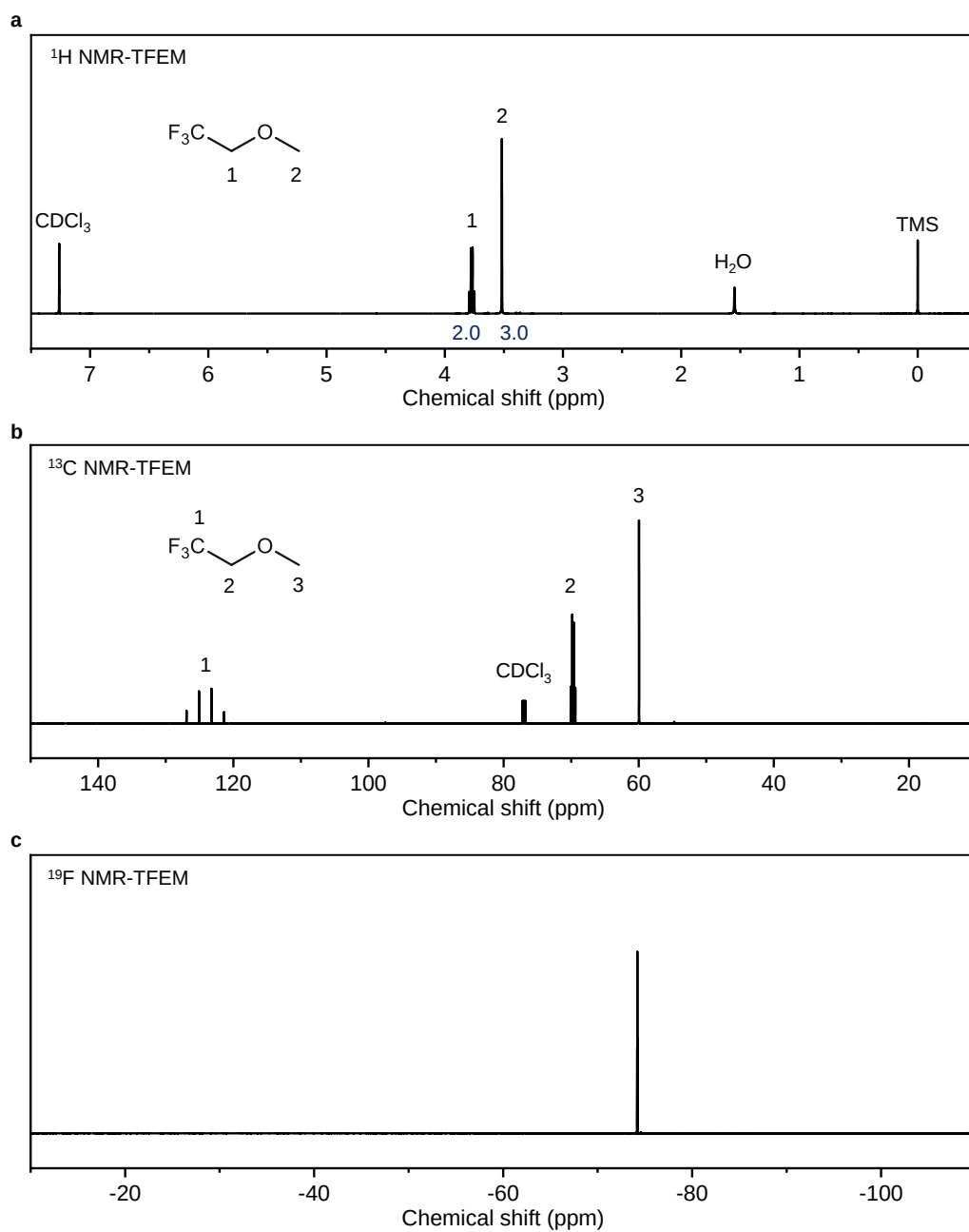
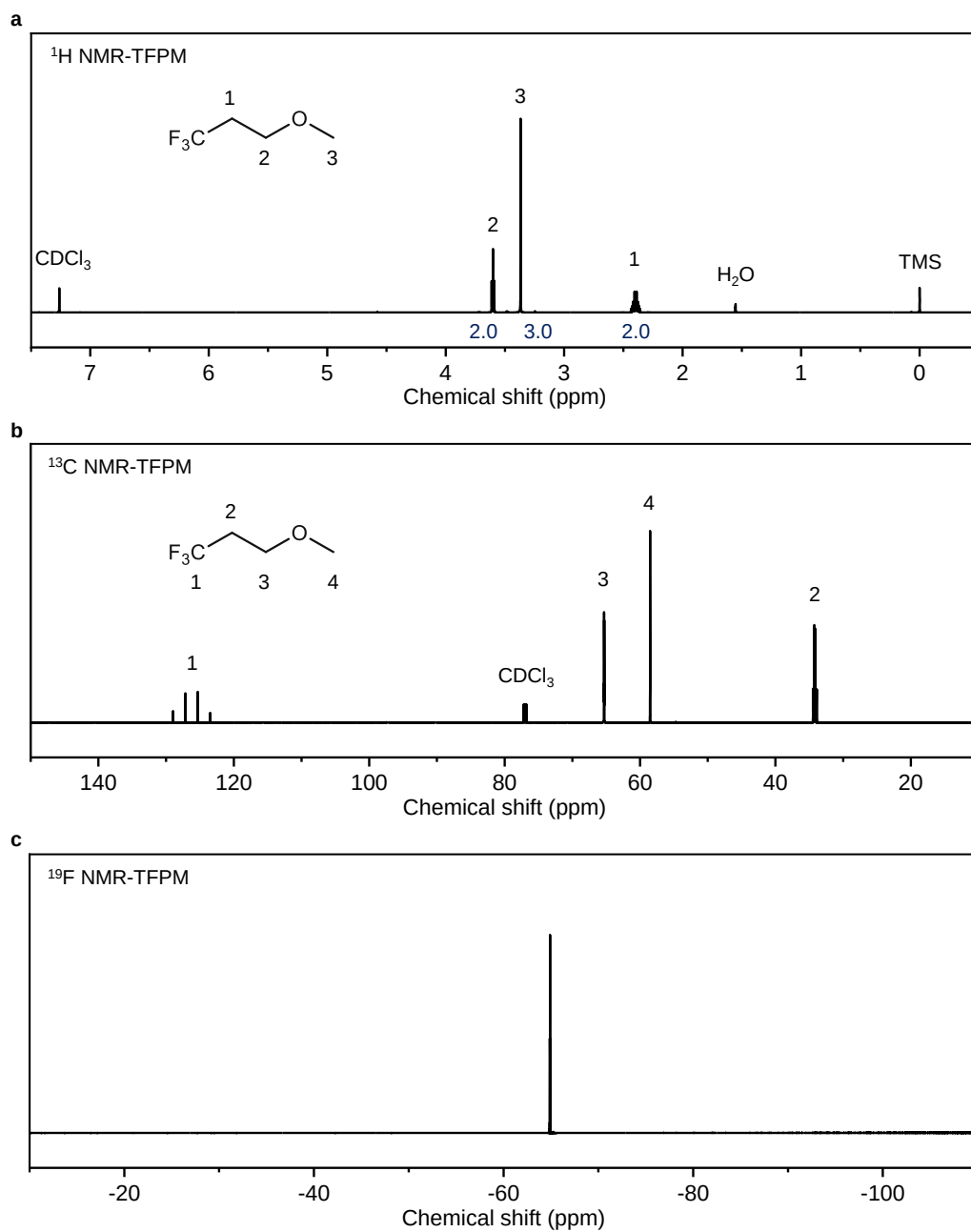




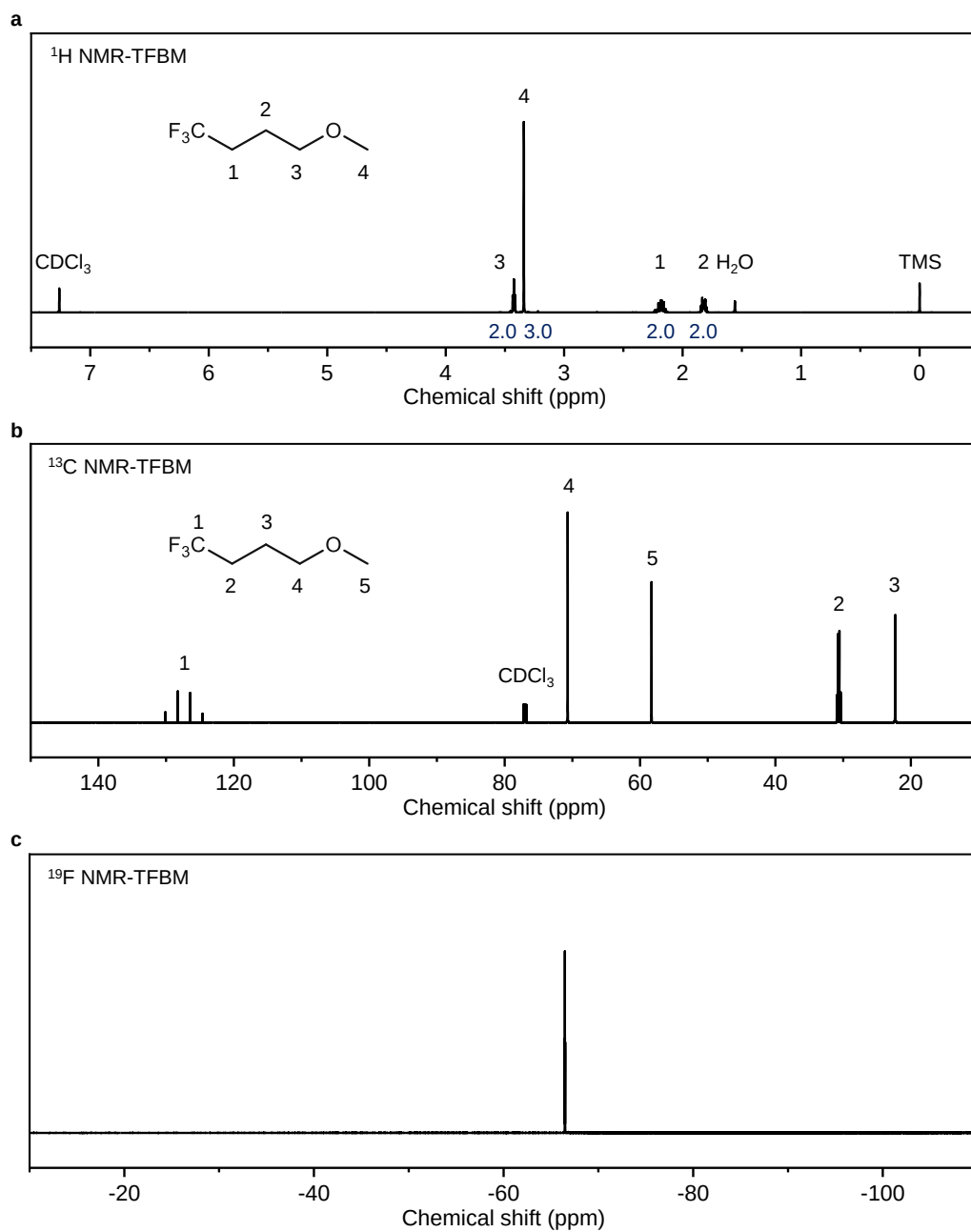
Supplementary Figure 1. Synthesis of the asymmetric fluorinated ethers.



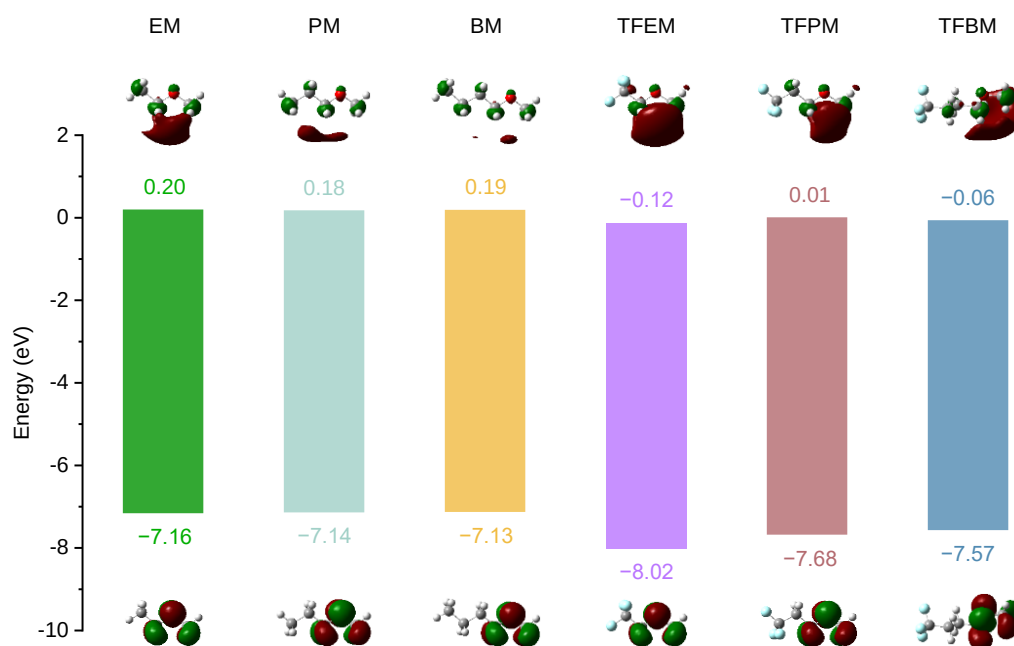
Supplementary Figure 2. NMR spectra of (a) ¹H, (b) ¹³C, and (c) ¹⁹F of TFEM in CDCl₃.



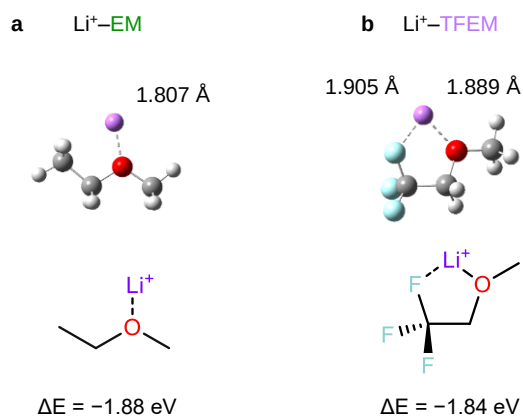
Supplementary Figure 3. NMR spectra of (a) ¹H, (b) ¹³C, and (c) ¹⁹F of TFPM in CDCl₃.



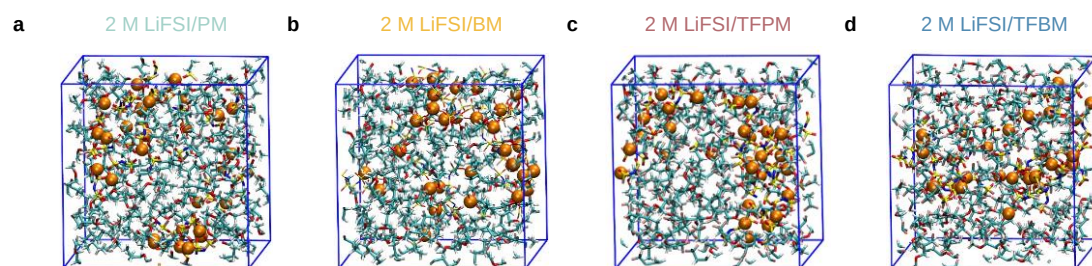
Supplementary Figure 4. NMR spectra of (a) ¹H, (b) ¹³C, and (c) ¹⁹F of TFBM in CDCl₃.



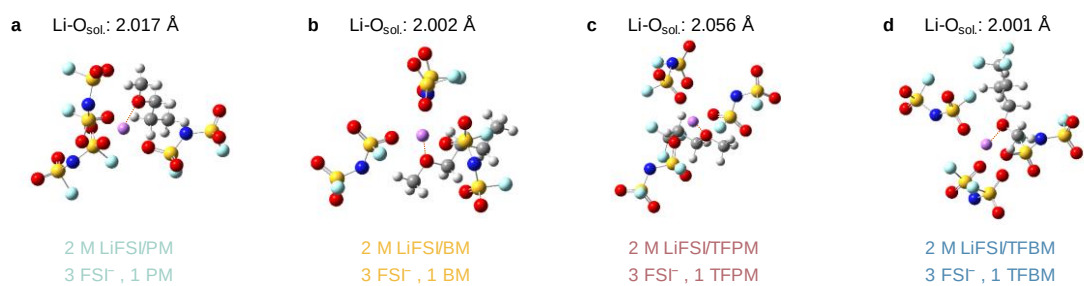
Supplementary Figure 5. Molecular orbital energies of solvents. Purple, red, cyan, grey, and white spheres represent Li^+ , O, F, C, and H, respectively. The LUMO is shown with the phase of the molecular orbital wavefunction indicated: red denotes positive phase and green denotes negative phase. Source data are provided as a Source Data file.



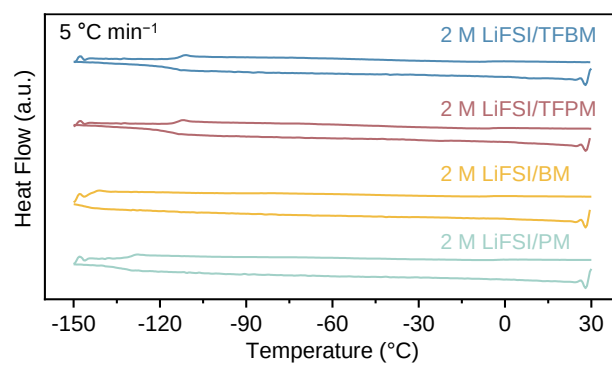
Supplementary Figure 6. Coordination structures and binding energies between one Li^+ and one solvent calculated from DFT. (a) $\text{Li}^+ \text{--EM}$. (b) $\text{Li}^+ \text{--TFEM}$. Purple, red, cyan, grey, and white spheres represent Li^+ , O, F, C, and H, respectively. Source data are provided as a Source Data file.



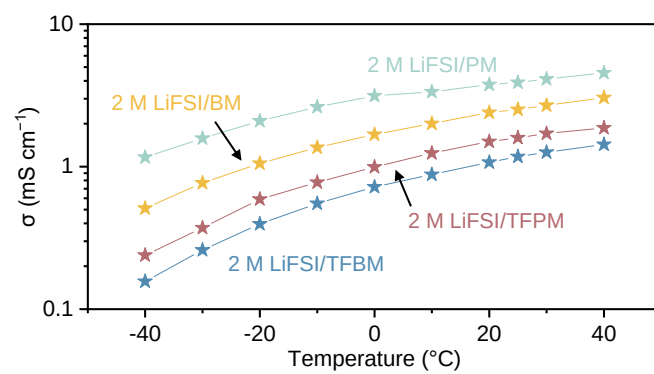
Supplementary Figure 7. Molecular dynamics (MD) simulation snapshots of (a) 2 M LiFSI/PM, (b) 2 M LiFSI/BM, (c) 2 M LiFSI/TFPM, and (d) 2 M LiFSI/TFBM electrolytes. Source data are provided as a Source Data file.



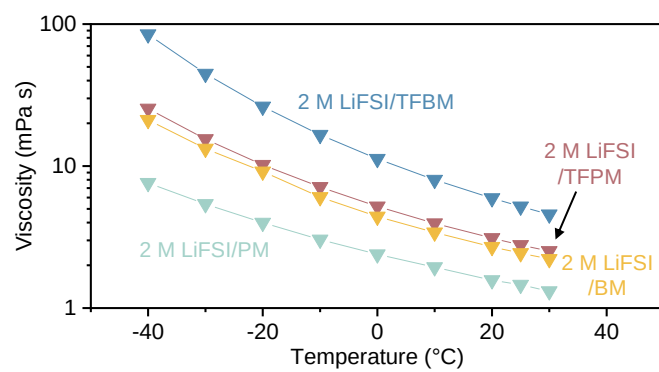
Supplementary Figure 8. The most probable solvation structures of (a) 2 M LiFSI/PM, (b) 2 M LiFSI/BM, (c) 2 M LiFSI/TFPM, and (d) 2 M LiFSI/TFBM electrolytes extracted from MD simulations. Purple, red, cyan, blue, yellow, grey, and white spheres represent Li⁺, O, F, N, S, C, and H, respectively. Source data are provided as a Source Data file.



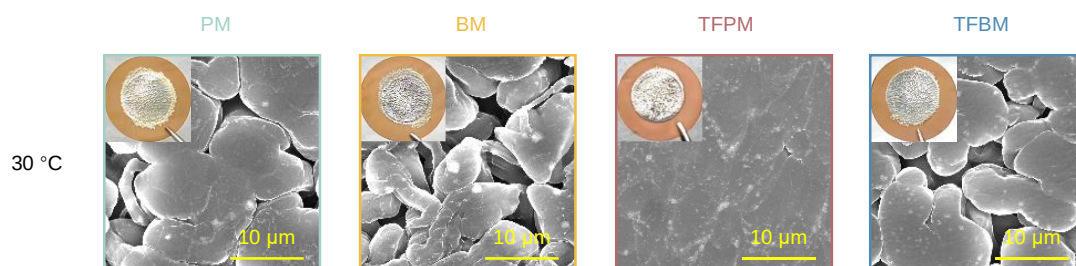
Supplementary Figure 9. DSC curves of different electrolytes with temperature range from -150 to 30 °C at a rate of 5 °C min⁻¹. Source data are provided as a Source Data file.



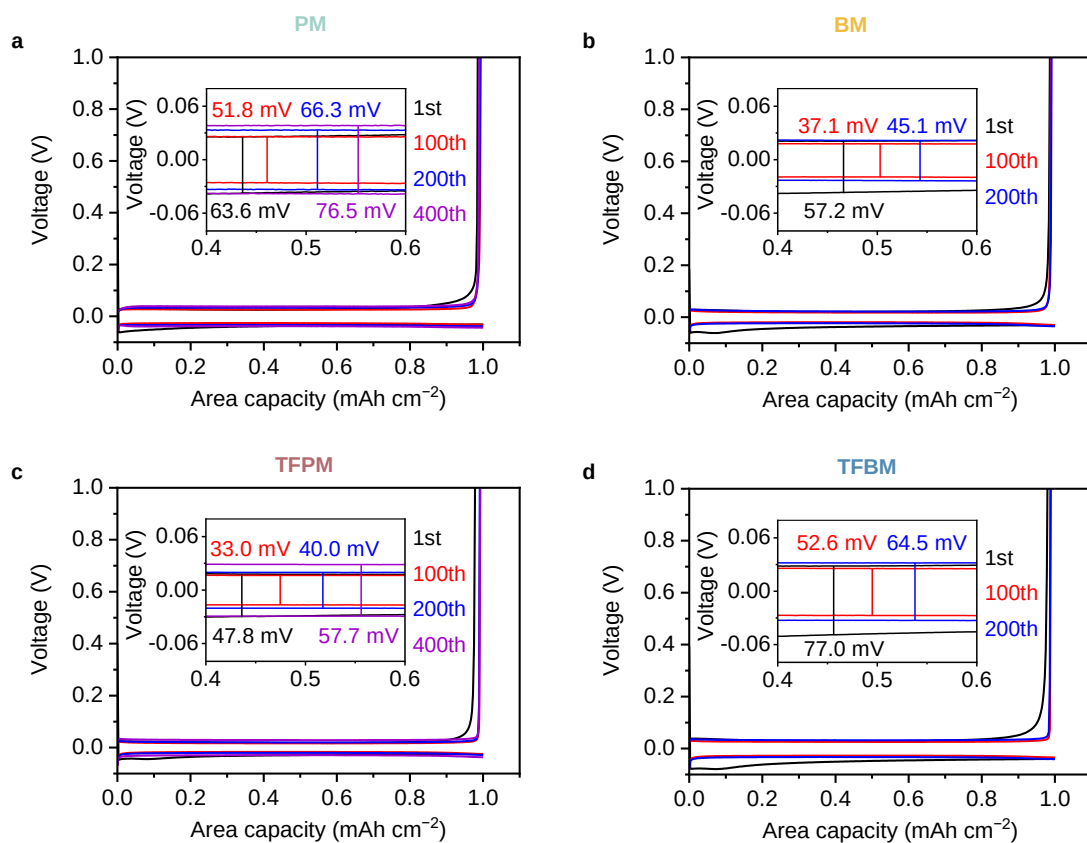
Supplementary Figure 10. Temperature dependent ionic conductivity of different electrolytes. Source data are provided as a Source Data file.



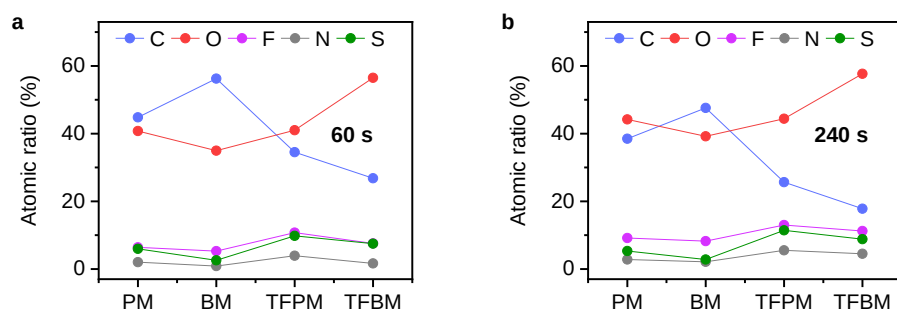
Supplementary Figure 11. Temperature dependent viscosity of different electrolytes. Source data are provided as a Source Data file.



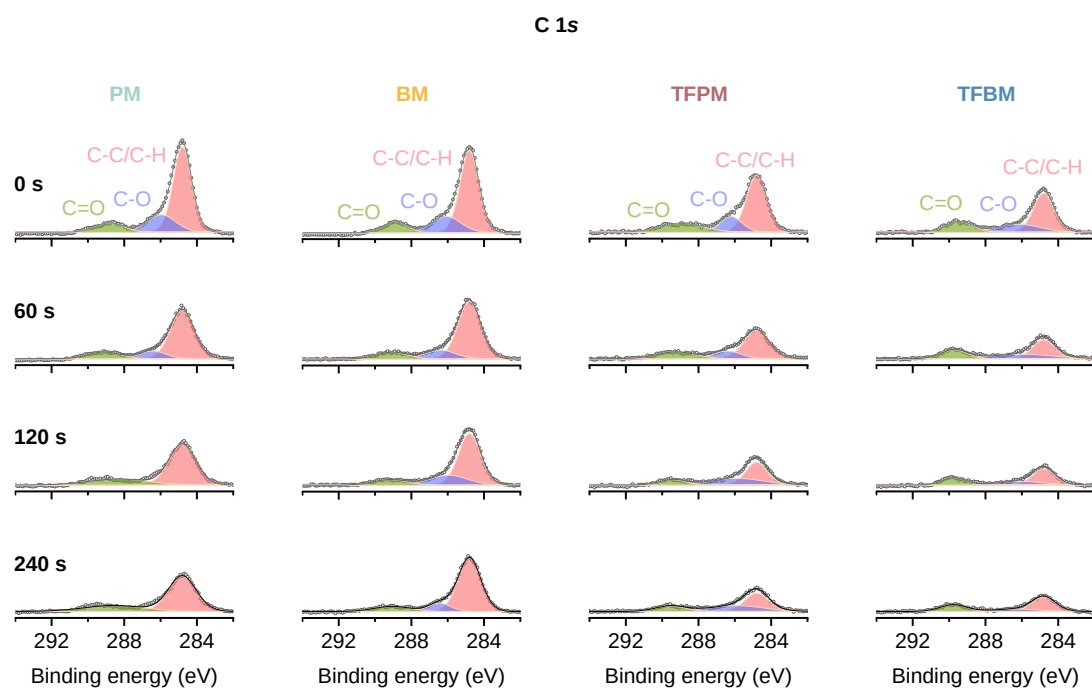
Supplementary Figure 12. SEM images and corresponding optical photographs of Cu foils using different electrolytes after plating 5.0 mAh cm^{-2} of Li metal at $30 \text{ }^{\circ}\text{C}$. (Scale bar = $10 \text{ }\mu\text{m}$).



Supplementary Figure 13. Corresponding Li plating/stripping profiles of Li||Cu cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/BM, (c) 2 M LiFSI/TFPM, and (d) 2 M LiFSI/TFBM electrolytes shown in Figure 4b. Source data are provided as a Source Data file.

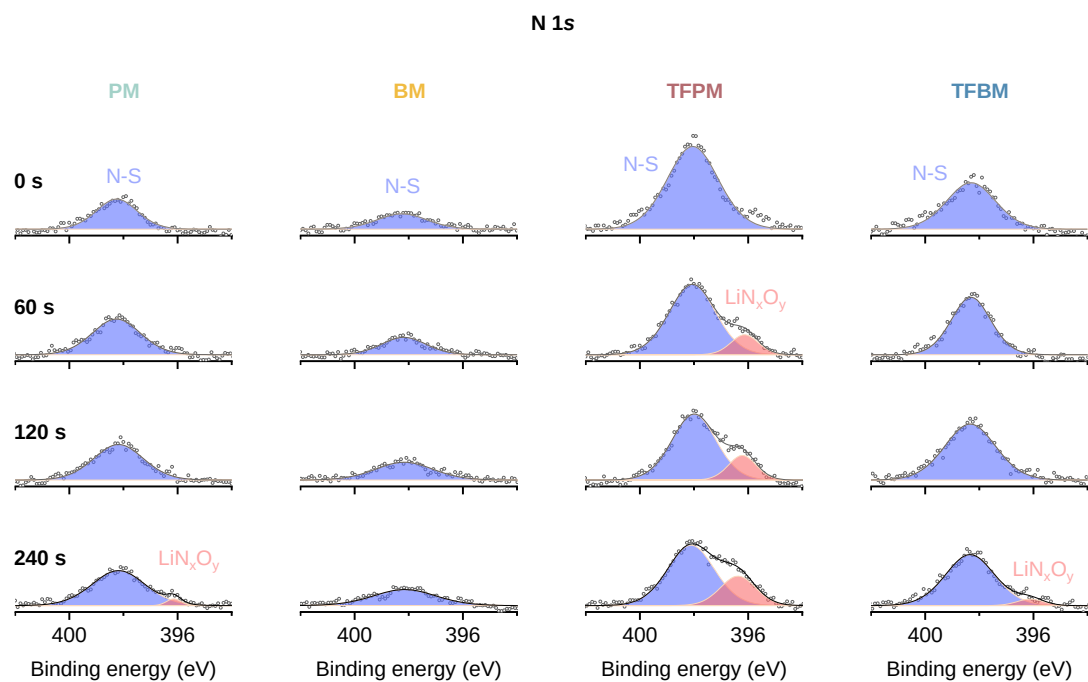


Supplementary Figure 14. Atomic ratio of the SEI on Li metal after (a) 60 s and (b) 240 s of sputtering. The Li metal plated on Cu foil was collected from Li||Cu cells after 10 cycles of Li plating/stripping in different electrolytes at 1.0 mA cm^{-2} for 1.0 mAh cm^{-2} at 30°C . Source data are provided as a Source Data file.



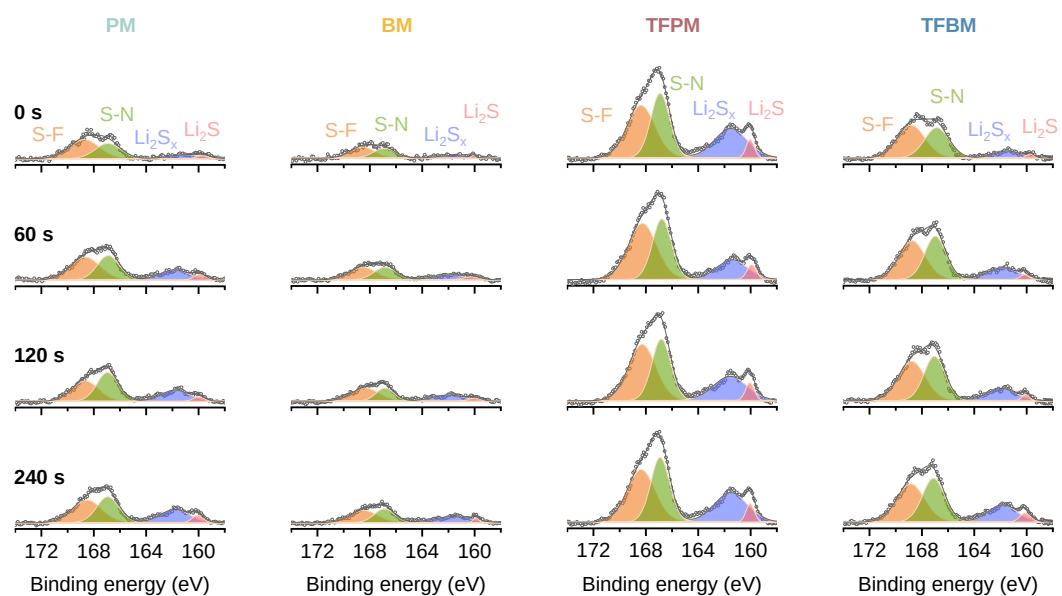
Supplementary Figure 15. C 1s XPS spectra of Li metal at different time of Ar⁺ sputtering.

Source data are provided as a Source Data file.

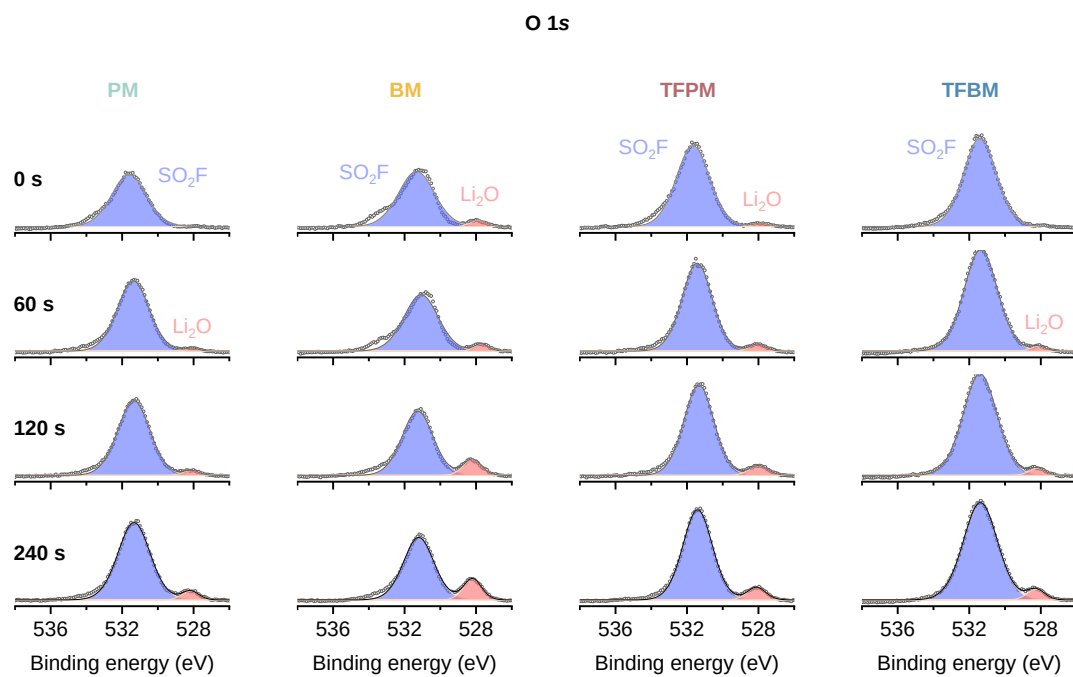


Supplementary Figure 16. N 1s XPS spectra of Li metal at different time of Ar⁺ sputtering. Source data are provided as a Source Data file.

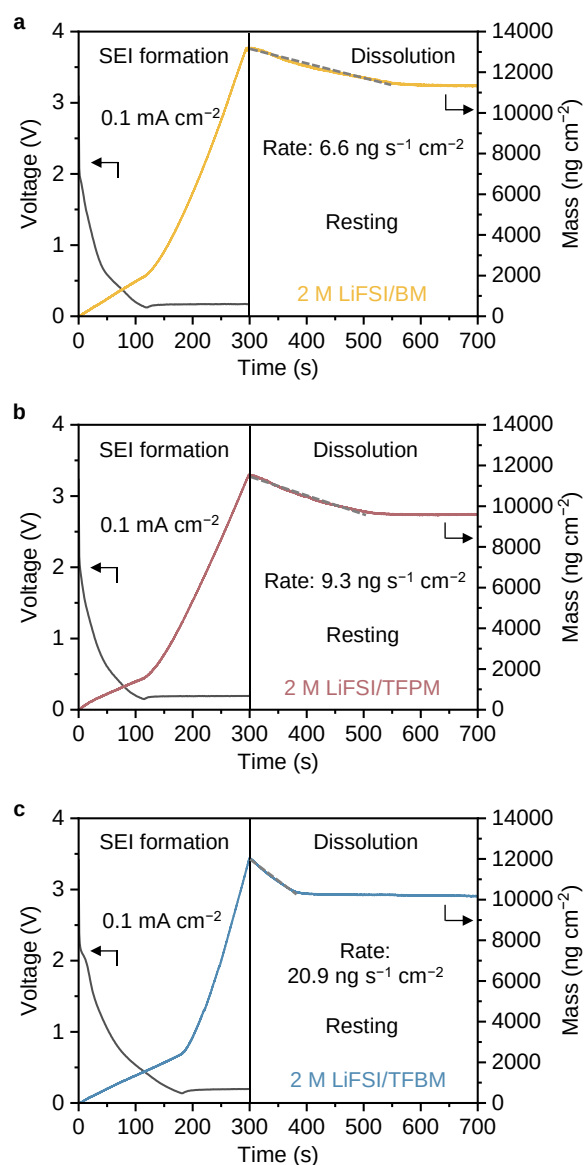
S 2p



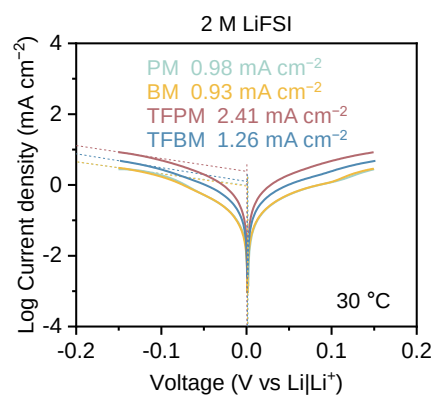
Supplementary Figure 17. S 2p XPS spectra of Li metal at different time of Ar^+ sputtering. Source data are provided as a Source Data file.



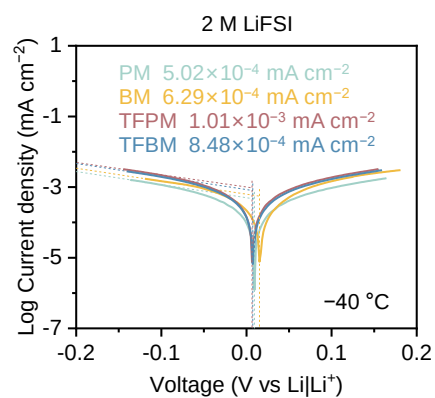
Supplementary Figure 18. O 1s XPS spectra of Li metal at different time of Ar^+ sputtering. Source data are provided as a Source Data file.



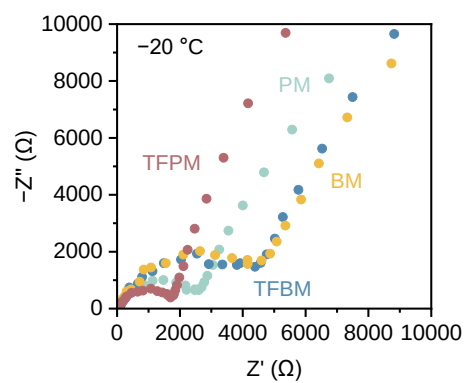
Supplementary Figure 19. In-situ EQCM results of mass change in (a) 2 M LiFSI/BM (low μ), (b) 2 M LiFSI/TFPM (moderate μ), and (c) 2 M LiFSI/TFBM (high μ) electrolytes during initial Li plating at a current density of 0.1 mA cm^{-2} and subsequent resting process. Source data are provided as a Source Data file.



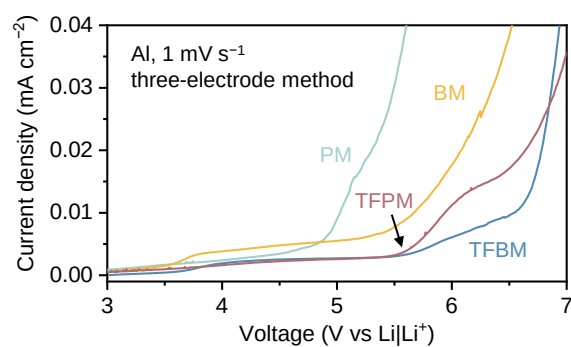
Supplementary Figure 20. Tafel plots of Li plating/stripping in four electrolytes at 30 °C. Source data are provided as a Source Data file.



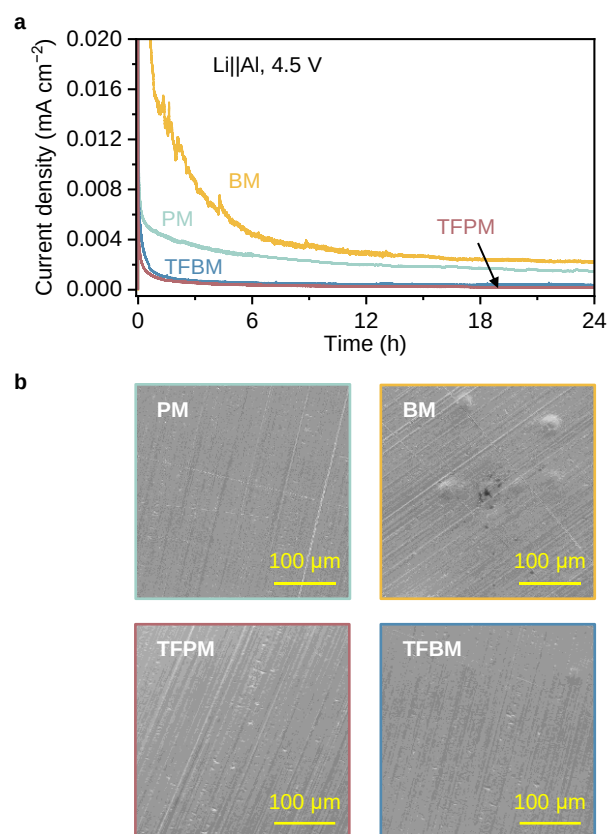
Supplementary Figure 21. Tafel plots of Li plating/stripping in four electrolytes at $-40\text{ }^{\circ}\text{C}$. Source data are provided as a Source Data file.



Supplementary Figure 22. EIS spectra of Li||Cu cells after 10 cycles with a current density of 1.0 mA cm^{-2} for 1.0 mAh cm^{-2} at $-20 \text{ }^{\circ}\text{C}$. Source data are provided as a Source Data file.

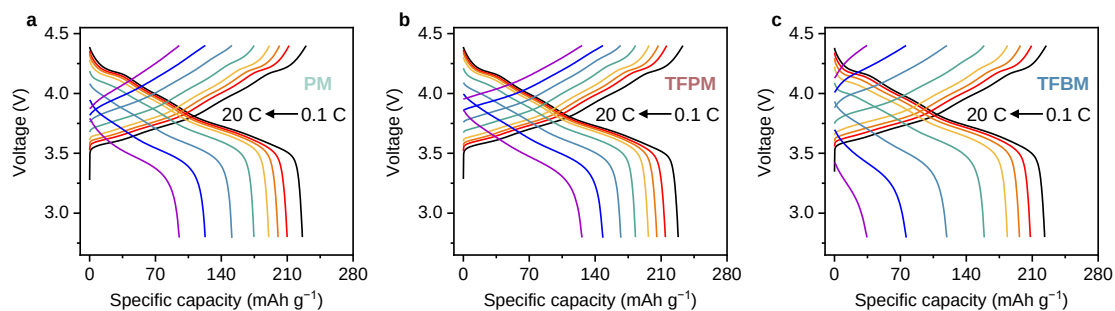


Supplementary Figure 23. LSV of four electrolytes in a three-electrode cell with Al as working electrode and Li as the counter and reference electrodes at a scan rate of 1 mV s⁻¹. Source data are provided as a Source Data file.



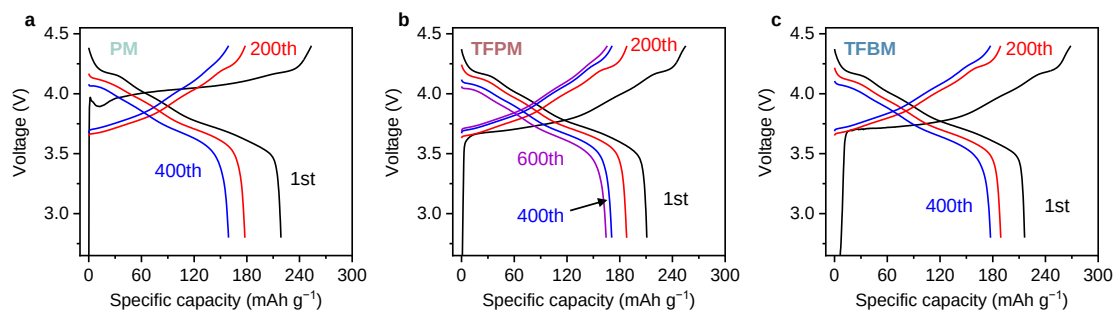
Supplementary Figure 24. (a) CA profiles of Li||Al cells at 4.5 V for 24 h in four electrolytes and (b) the SEM images of corresponding Al electrodes after CA test (scale bar = 100 μm). Source data are provided as a Source Data file.

30 °C, 4.4 V, Li||0.74 mAh cm⁻² NMC811



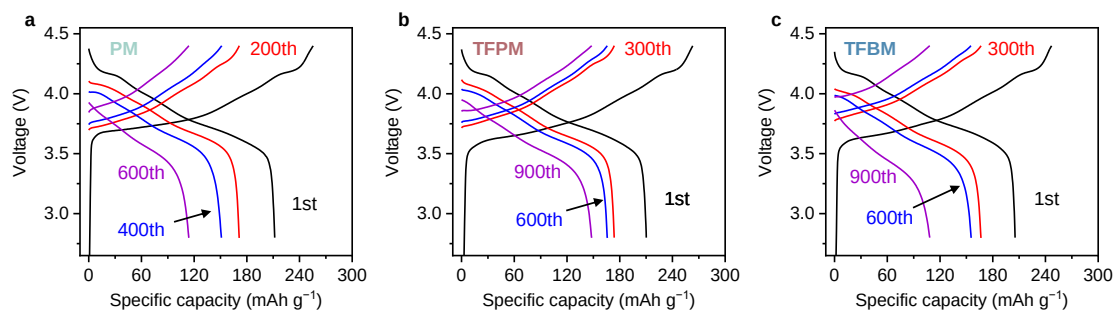
Supplementary Figure 25. Selected charge-discharge profiles of Li||NMC811 cells using (a) PM, (b) TFPM, and (c) TFBM electrolytes at different rates. Source data are provided as a Source Data file.

30 °C, 4.4 V, 2 C, Li||0.74 mAh cm⁻² NMC811

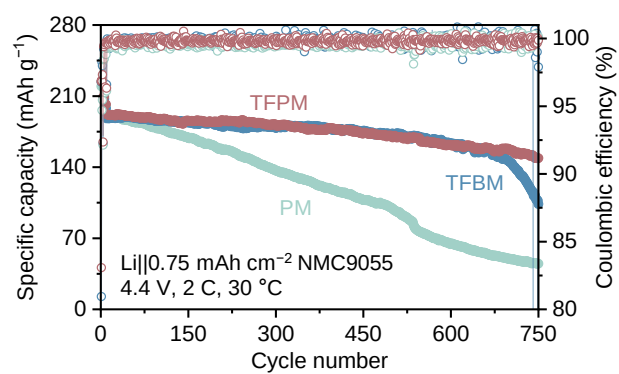


Supplementary Figure 26. Corresponding charge-discharge profiles of Li||NMC811 cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/TFPM, and (c) 2 M LiFSI/TFBM electrolytes shown in Figure 6c. Source data are provided as a Source Data file.

30 °C, 4.4 V, 5 C, Li||0.74 mAh cm⁻² NMC811

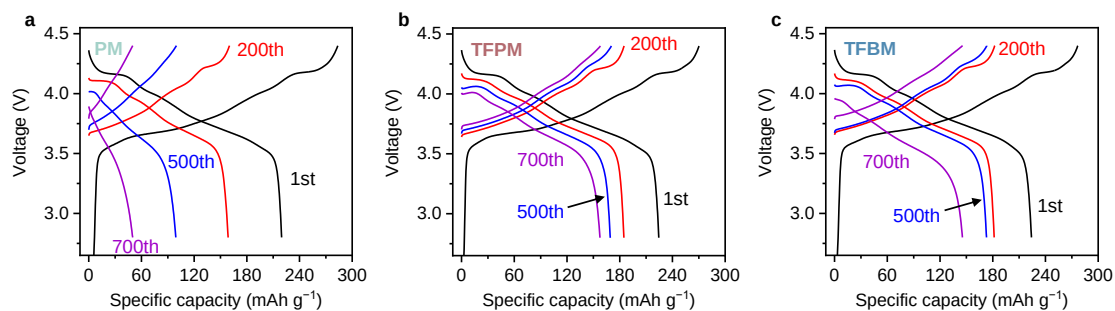


Supplementary Figure 27. Corresponding charge-discharge profiles of Li||NMC811 cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/TFPM, and (c) 2 M LiFSI/TFBM electrolytes shown in Figure 6d. Source data are provided as a Source Data file.



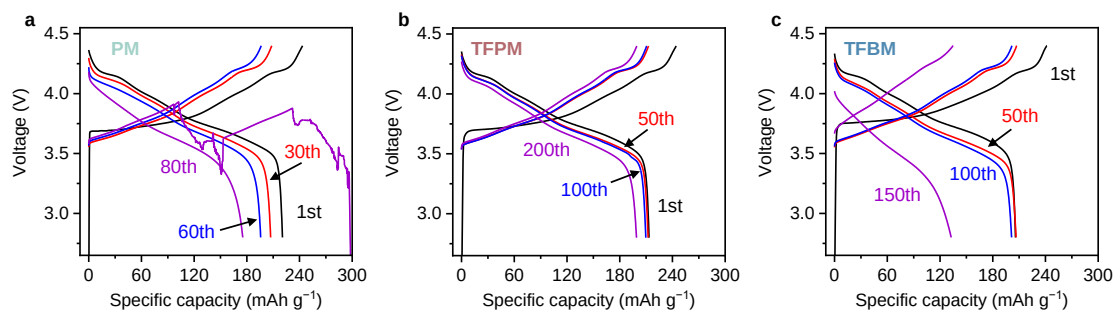
Supplementary Figure 28. Cycling performance of Li||NMC9055 cells with different electrolytes at 30 °C, 1 C = 215 mA g⁻¹. Source data are provided as a Source Data file.

30 °C, 4.4 V, 2 C, Li||0.75 mAh cm⁻² NMC9055



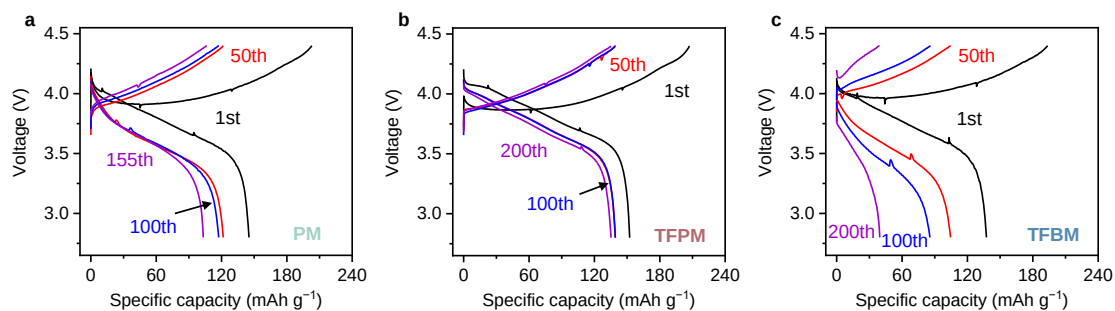
Supplementary Figure 29. Corresponding charge-discharge profiles of Li||NMC9055 cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/TFPM, and (c) 2 M LiFSI/TFBM electrolytes at 30 °C shown in Supplementary Figure 28. Source data are provided as a Source Data file.

30 °C, 4.4 V, 0.2 C / 0.5 C, 50 μm Li||4.0 mAh cm^{-2} NMC811

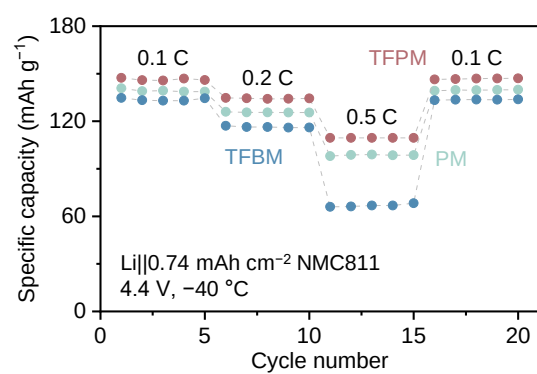


Supplementary Figure 30. Corresponding charge-discharge profiles of 50 μm Li||4.0 mAh cm^{-2} NMC811 cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/TFPM, and (c) 2 M LiFSI/TFBM electrolytes at 30 °C shown in Figure 6e. Source data are provided as a Source Data file.

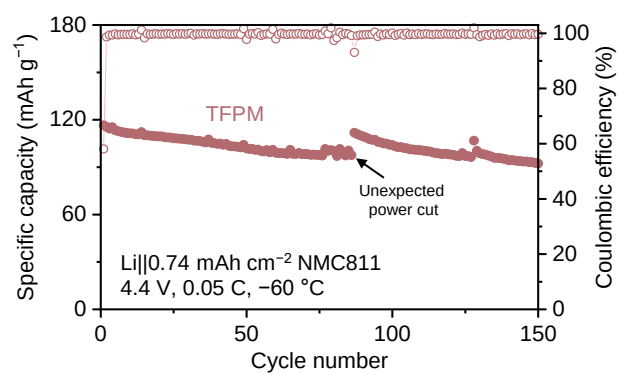
-40 °C, 4.4 V, 0.1 C, 50 μm Li||4.0 mAh cm^{-2} NMC811



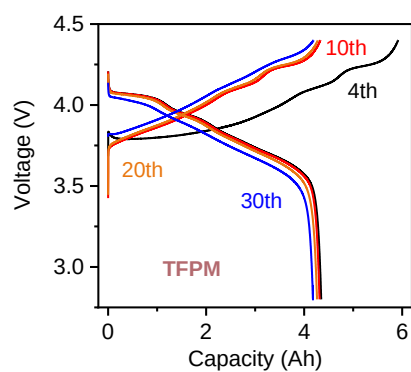
Supplementary Figure 31. Corresponding charge-discharge profiles of 50 μm Li||4.0 mAh cm^{-2} NMC811 cells with (a) 2 M LiFSI/PM, (b) 2 M LiFSI/TFPM, and (c) 2 M LiFSI/TFBM electrolytes at -40 °C shown in Figure 6f. Source data are provided as a Source Data file.



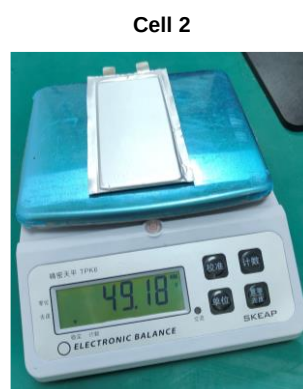
Supplementary Figure 32. Rate performance of the Li||NMC811 cells at -40 °C. Source data are provided as a Source Data file.



Supplementary Figure 33. Cycling performance of Li||NMC811 cell with TFPM electrolyte at -60 °C. Source data are provided as a Source Data file.



Supplementary Figure 34. Selected charge-discharge profiles of the Li||NMC9055 pouch cell with the TFPM electrolyte at $-40\text{ }^{\circ}\text{C}$. Source data are provided as a Source Data file.



Supplementary Figure 35. Weights of two pouch cells.

Supplementary Table 1. Physicochemical properties of the solvents.

	Molecular weight (g mol⁻¹)	Density (g cm⁻³, 25 °C)	Viscosity (mPa s, 25 °C)	Boiling point (°C, 1 atm)
EM	60.10	0.72510 (0 °C)	/	7
PM	74.12	0.71997	0.1983	39
BM	88.15	0.73927	0.2883	71
TFEM	114.07	1.16610 (3 °C)	/	32
TFPM	128.09	1.10376	0.3768	54
TFBM	142.12	1.04910	0.4457	74

Supplementary Table 2. The Li||Cu CE comparisons of this work with state-of-the-art reported single-salt single-solvent electrolytes at a current density of 1.0 mA cm⁻².

Electrolyte	Current density (mA cm ⁻²)	Capacity (mAh cm ⁻²)	Cycle number	CE (%)	Ref.
2 M LiFSI/TFPM	1	1	850	99.3 (20-850 cycle)	This work
2 M LiFSI/BFE	1	1	300	/	[1]
1 M LiFSI/cFTOF	1	1	200	97.9	[2]
1 M LiFSI/DTD L	1	1	300	98.2 (100-300 cycle)	[3]
2 M LiFSI/TFDOL	1	1	120	98.5	[4]
2 M LiFSI/DMP	1	1	200	98.5	[5]
1 M LiFSI/BFME	1	1	450	98.8	[6]
1 M LiFSI/DMM	1	1	100	98.8	[7]
1 M LiFSI/DEM	1	1	250	99.1	[8]
2 M LiFSI/TFDMP	1	1	450	99.1	[9]
3 M LiFSI/FMS	1	1	490	99.1	[10]
1.6 M LiFSI/AzTFSA	1	1	200	99.2	[11]
1.5 M LiFSI/NMBE	1	1	200	99.3	[12]

Supplementary Table 3. The Li||Cu CE comparisons of this work with state-of-the-art reported electrolytes at low temperatures (≤ -40 °C).

Electrolyte	Temperature (°C)	Current density/Capacity (mA cm ⁻² /mAh cm ⁻²)	CE (%)	Ref.
2 M LiFSI/TFPM	-40	1.0-1.0	98.7	This work
	-60	0.1-0.1	96.1	
1 M LiFSI/EMC-FEC-DTF (1.5:1.5:7 by vol)	-40	0.25-0.5	94.2	[13]
1.5 M LiFSI/DMS	-40	0.25-0.25	94.52	[14]
1 M LiFSI/DMM	-40	0.5-0.5	97.87	[7]
2 M LiFSI/DBE	-40	0.5-1.0	98.2	[15]
0.75 M LiFSI/TFMSA, TFMSA/TTE (7:3 by vol)	-40	0.33-1.0	98.5	[16]
1 M LiFSI/TFMP-DME (7:1 by vol)	-40	1.0-1.0	98.5	[17]
1 M LiClO ₄ /ES-10 wt% FEC	-50	0.5-1.0	93.33	[18]
LiFSI-FSNDM (1:5 by mol)	-50	0.5-1.0	96.94	[19]
1 M LiFSI/BTFE-DME (5:1 by vol)	-40	0.25-1.0	98.5	[20]
	-60	0.25-1.0	96.9	
1 M LiFSI/DEE	-40	0.5-1.0	99.0	[21]
	-60	0.5-1.0	98.4	
LiFSI-MPE-TFME (1:2:4.5 by mol)	-40	0.5-1.0	99.34	[22]
	-60		98.96	
LiFSI-DEE-BTFE (1:2.6:2 by mol)	-40	0.5-1.0	99.1	[23]
	-60	0.25-1.0	98.9	

Supplementary Table 4. Performance comparison this work with state-of-the-art reported Li-metal coin-cells with high areal capacity and low N/P ratio at low temperatures (≤ -20 °C).

Electrolyte	Temperature (°C)	Li thickness (μm)	Positive electrode	Areal Capacity (mAh cm^{-2})	N/P ratio	Cycling retention	CE (%)	Ref.
2 M LiFSI/TFPM	-40	50	NMC811	4.0	2.6	94.1%@200cycles	99.9	This work
1 M LiFSI-0.2 M LiTfD/DME	-20	40	NMC811	2.6	4	81.2%@200cycles	99.8	[24]
1 M LiClO ₄ /ES-10 wt% FEC	-33	/	NMC811	2.0	5	~80%@100cycles	/	[18]
2 M LiFSI/BFE	-30	40	NMC811	3.5	2.8	>90%@180cycles	/	[1]
1 M LiFSI-0.3M LiNO ₃ /THF	-40	40	NMC811	4.2	2.31	68%@80cycles	/	[25]
1.5 M LiFSI/DMS	-40	50	NMC811	4.3	2.32	~95%@35cycles	/	[14]
1 M LiFSI/BTFE-DME (5:1 by vol)	-40	4 mAh cm ⁻²	NMC811	2.1	2	~100%@200cycles	/	[20]
LiFSI-FSNDM (1:5 by mol, ~1.5 M)	-20	50	NMC811	~4.70	~2.13	only 6cycles	/	[19]
	-50			~1.72	~5.81			
1 M LiFSI/THF + 0.5 wt% NaPFO	-60	40	NMC811	2.6	3	72%@80cycles	/	[26]
1 M LiFSI/DEE-DME 9:1 vol% + 0.1 M PN	-60	40	NMC811	3.0	3	93.3%@110cycles	/	[27]
1.5 M LiFSI/THF-2MeTHF-TTE (1:1:2 by mol)	-20	50	LFP	1.5	6	~100%@150cycles	/	[28]
	-40							
1 M LiFSI/DMM	-20	50	LTO	2.64	2.78	75.9%@100cycles	/	[7]
LiFSI-DEE-BTfE (1:2.6:2 by mol)	-40	20	NCA	2	2.3	95%@100cycles	/	[23]
	-60	20				93%@100cycles		

0.75 M LiFSI/TFMSA, TFMSA/TTE (7:3 by vol)	-40	50	LCO	4.0	2.5	50%@25cycles	99.7	[16]
LiFSI-MPE-TFME (1:2.3:4.5 by mol)	-40	50	LCO	2.7	/	90%@200cycles	/	[22]
	-60					97%@50cycles		

Supplementary Table 5. Detailed parameters of the Li||NMC9055 pouch cell (cell 1).

Material	Parameter	Value
Positive electrode(NMC9055)	Mass loading (one side)	19 mg cm ⁻²
	Active material content	95.5%
	Layers	8
Negative electrode (Li-metal)	Thickness (one side)	40 μm
	Layers	9
	N/P ratio	1.0
Electrolyte	Weight	11.75
	E/C ratio	1.89 g Ah ⁻¹
Full cell	Dimension	12.7 × 5.6 cm ²
	Pressure	100 kPa
	Capacity	6.22 A h
	Mid-value voltage	3.84 V
	Total weight	48.94 g
	Specific energy (25 °C)	488.0 Wh kg ⁻¹

Supplementary Table 6. Detailed parameters of the Li||NMC9055 pouch cell (cell 2).

Material	Parameter	Value
Positive electrode (NMC9055)	Mass loading (one side)	19 mg cm ⁻²
	Active material content	95.5%
	Layers	8
Negative electrode (Li-metal)	Thickness (one side)	40 μm
	Layers	9
	N/P ratio	1.0
Electrolyte	Weight	11.77 g
	E/C ratio	1.87 g Ah ⁻¹
Full cell	Dimension	12.7 × 5.6 cm ²
	Pressure	100 kPa
	Capacity (25 °C)	6.29 Ah
	Capacity (-40 °C)	4.40 Ah
	Mid-value voltage (25 °C)	3.84 V
	Mid-value voltage (-40 °C)	3.86 V
	Total weight	49.18 g
	Specific energy (25 °C)	491.1 Wh kg ⁻¹
	Specific energy (-40 °C)	345.3 Wh kg ⁻¹

Supplementary Table 7. Performance comparison this work with state-of-the-art reported Li-metal pouch cells at low temperatures (≤ -20 °C).

Electrolyte	Temperature (°C)	Capacity (Ah)		Specific energy (Wh kg ⁻¹)		Cathode	Cut-off voltage (V)	N/P ratio	E/C ratio (g Ah ⁻¹)	Cycling retention	Rate (cha/dis) (C)	Ref.
		RT	LT	RT	LT							
2 M LiFSI/TFPM	-40	6.29	4.40	491.1	345.3	NMC9055	4.4	1.0	1.87	90.8%@40cycles	~0.05/~0.05 (310 mA)	This work
1 M LiFSI-3 wt% LiNO ₃ /THP-5 vol% FEC	-20	2.7	1.98	403	340	NMC811	4.5	/	2.2	88%@40cycles	0.2/0.2	[29]
1.5 M LiFSI/DMS	-20	1.2	0.98	/		NMC811	4.3	/	~2.5	92.1%@50cycles	0.05/0.1	[14]
1 M LiPF ₆ -0.5 wt% AIBN-1.5 wt% PETEA/TFEA-EA 2F-FEC (1:7.2:1.8 by vol)	-20	2.90	~2.26	450	350	NMC811	4.4	/	1.55	83.4%@75cycle	0.05/0.1	[30]
1 M LiFSI/EMC- FEC-DTF (1.5:1.5:7 by vol)	-40	2.0	1.71	/		NMC811	4.8	1.2	2.8	~100%@20cycles	0.067/0.2	[13]
1 M LiFSI-0.5 wt% NaPFO-2 wt% TMSP/THF	-40	~0.50	0.37	~442*	327*	NMC811	4.3	1.26	2.5	< 70%@20cycles	0.1/0.1	[26]
1.5 M LiFSI/THF- 2MeTHF-TTE (1:1:2 by mol)	-20	1.06	0.69	/		LFP	4.0	3	3	~100%@24cycles	0.3/0.3	[28]
2.1 M LiTFSI/MTBE-THF (3:1 by vol)	-40	1.2	> 0.76	/		LFP	3.8	3.6	3	~100%@87cycles	0.2/0.2	[31]

Note: * represents excluding the weight of tabs and packing foil when calculating the specific energy.

Supplementary Table 8. Rationale for radar plot.

Aspect	Reference data & ratings	2 M LiFSI/solvent			
		PM	BM	TFPM	TFBM
Anodic stability	Leakage current in CA test of Li Al cells after 24 h ($\mu\text{A cm}^{-2}$)	1.48	2.22	0.19	0.34
	Ratings	6	4	10	9
Li-metal compatibility	Avg. CE (%) and cycle number of Li Cu cells	99.2@400	99.1@260	99.3@850	99.0@380
	Ratings	8	7	9	7
De-solvation	Avg. ΔE_{dsv} (eV)	2.25	2.62	2.14	2.29
	Ratings	8	7	9	8
Ionic conduction	Ionic conductivity at 30 °C (mS cm^{-1})	4.12	2.69	1.71	1.26
	Ratings	9	8	7	6
Interphase stability	The S and F ratios (%) in XPS after 120 s of Ar^+ sputtering, indicative of robust inorganic components in the SEI.	4.84	2.51	11.44	8.38
		9.48	6.05	12.60	8.85
	Ratings	7	5	9	7

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